

PROJECTS

LLM ARCHITECTURE EVOLUTION [GitHub](#)

Built and analyzed 5 language models representing major architectural shifts in nearly a decade of Transformer research, training them across **10B+ tokens** at 75M active parameters.

- Reduced held-out perplexity by **47.6%** across the five-model benchmark under identical evaluation conditions.
- Improved content-token top-1 accuracy by **44.1%** and top-10 accuracy by **28.9%** through controlled architecture comparisons.
- Reduced content cross-entropy by **22.1%** and extreme rank>1000 prediction failures by **56.3%**.
- Quantified quality-efficiency trade-offs across architectures, revealing up to **13x** differences in inference speed and **4.6x** in training throughput.

LIGHTWEIGHT SENTENCE EMBEDDINGS [GitHub](#)

Research and engineering project combining a custom lightweight sentence-embedding model with an internal company knowledge assistant.

- Designed a 1.32 MB custom sentence encoder achieving 0.777 Pearson correlation, within 0.05 of MiniLM while being **65x smaller and 210x faster** in forward inference.
- Built a FastAPI + Supabase knowledge assistant that routes queries through semantic FAQ matching, document-based RAG, and confidence-based rejection using the custom embedding model.
- Reduced unnecessary LLM usage versus a conventional always-RAG pipeline, serving FAQ and out-of-domain queries in **1.5–3.2 ms vs. 1.67 s** for RAG, by routing requests through FAQ matching, confidence thresholds, and semantic caching before invoking the LLM.

EXPERIENCE

FULLSTACK DEVELOPER (VOLUNTEER)

Fundacja Centrum Św. Ojca Pio (Feb 2026 – present)

- Developed a responsive Next.js + TypeScript + Tailwind CSS website with multiple content pages and accessibility best practices.
- Built a custom admin panel with server-side authentication and CRUD management for website content, team members and activities.
- Implemented a lightweight Next.js backend with persistent content storage, Google Maps integration and self-hosting support.

JUNIOR FULLSTACK DEVELOPER

Jobsdoor (Jul 2024 – Oct 2024)

- Developed and enhanced features of a job portal - from simple UI components (e.g., banners) to key modules like the pricing page
- Implemented automatic translations for the website and admin panel (articles and dynamic content)
- Worked with technologies: TypeScript, React, Go, PostgreSQL, Docker

AUTOMATION DEVELOPER

Jidoka (Feb 2024 – Jul 2024)

- Automated LinkedIn and email messaging campaigns using Python and JavaScript scripts
- Wrote web scraping scripts with Python libraries such as BeautifulSoup and requests
- Worked in a team following the Scrum methodology

SKILLS

PROGRAMMING LANGUAGES

- Python
- SQL
- Javascript
- Typescript
- HTML
- CSS

TOOLS

- Git
- Docker
- Jupyter Notebook
- Google Cloud
- Linux

EDUCATION

LODZ UNIVERSITY OF TECHNOLOGY

Information Technology (English-taught)
Oct 2022 - Jun 2026

THE HAGUE UNIVERSITY OF APPLIED SCIENCES

Becoming the next successful online start-up (Erasmus exchange)
Feb 2025 - Jul 2025

ACHIEVEMENTS

- [Huawei Tech Arena Poland 2025](#)
- 1st place. Project: Custom architecture for low latency sentence embeddings
- [HACKNATION 2025](#) - finalist.
Project: Data anonymization and synthetic data generation
- [Collabothon 2024](#) – 1st place.
Project: ESG widget for corporate banking clients
- [Hack na Zdrowie 2](#) – 1st place.
Project: Onboarding platform with RAG AI chatbot

FINISHED COURSES

- [Neural Networks and Deep Learning](#)
- [Hyperparameter Tuning, Regularization and Optimization](#)
- [Structuring Machine Learning Projects](#)
- [Convolutional Neural Networks](#)

LANGUAGES

- Polish – Native
- English – C1